# HARIRAMAKRRISHNAN RAMACHANDRAN

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## PROFESSIONAL EXPERIENCE — 3.5 YEARS

**Software Engineer — HCL Technologies, Chennai, India** Sep 2021 – Jan 2025

- Designed and implemented **Python**-based data pipelines on **AWS Lambda** to process and transform large datasets for AI model training, handling approximately **3,500 events/sec** during peak hours.
- Developed and maintained **Java Spring Boot** microservices for AI infrastructure, focusing on building scalable RESTful APIs to serve model inference requests and manage data streams.
- Engineered **SQL** queries and managed schema evolution for **PostgreSQL** databases, ensuring efficient data retrieval and integrity for AI training datasets and operational logging.
- Containerized backend services using **Docker** and deployed them to a managed **Kubernetes** cluster (EKS), establishing CI/CD pipelines with **Jenkins** for automated testing and deployment.
- Optimized system performance and reliability for AI workloads by implementing caching strategies and refining database interactions, reducing average response times by **18%**.
- Collaborated asynchronously with a distributed team to define API contracts and integrate backend services with third-party data sources, documenting all architectural decisions in **Confluence**.
- Troubleshot and resolved production issues in a high-throughput environment, analyzing logs from **Splunk** and **CloudWatch** to identify root causes and implement preventative measures.

## SKILLS

**Languages** – Python, Java, SQL, Bash

**Frameworks & Libraries** – Spring Boot, FastAPI, pandas, NumPy, scikit-learn

**Databases** – PostgreSQL, MySQL, Oracle, MongoDB, Redis

**Cloud & DevOps** – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes, Jenkins, Git

**AI & ML Infrastructure** – Model Serving, Data Pipelines, ML Training Workloads, Inference Workloads, Performance Optimization

**API & Architecture** – RESTful API Design, Microservices, Distributed Systems

## PROJECTS

### Scalable ML Inference Service

- Developed a **FastAPI** service in **Python** to host a pre-trained machine learning model for real-time anomaly detection, achieving a p95 latency of **180ms**.
- Integrated the inference service with **PostgreSQL** for feature retrieval and **Redis** for caching, optimizing data access patterns to support high-volume requests.
- Containerized the service using **Docker** and deployed it to an **AWS EKS** cluster, establishing a CI/CD pipeline with **Jenkins** for automated builds and deployments.

### AI Data Pipeline Orchestration

- Engineered a data pipeline using **Python** and **AWS Lambda** to ingest and preprocess data from various sources, transforming approximately **1.5TB** of raw logs weekly.
- Implemented data validation checks and error handling mechanisms to ensure data quality for downstream machine learning model training, reducing data-related training failures by **30%**.
- Configured **Airflow** to orchestrate the data pipeline, scheduling and monitoring job execution to maintain reliable data flow to the feature store.

## EDUCATION

**M.Sc. Data Analytics — National College of Ireland, Dublin** Feb 2026

**B.E. Computer Science — SNS College of Technology, Coimbatore, India** 2021

## CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)